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Crypto Trend Prediction Based on Wavelet Transform and Deep Learning Algorithm

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Abstract

Crypto currency price prediction is one of the biggest challenges for any prediction system owing to its volatility and dependence on many factors. Artificial intelligence powered by the development of sophisticated computing technology and developments in research has opened up a wide variety of applications and new possibilities to solve complicated problems. Extraction of appropriate features and input with features affecting the output is equally important in time series prediction problems. Wavelet transform is considered an effective method to split a non-stationary signal into uncorrelated components. This work attempts to extract the advantage of wavelets to extract features from an available data set and to predict the trend of the crypto price for next 16 days using the Bi-LSTM network. Historic data of crypto currencies available in Yahoo finance through the yFinance API with the python code is used as the primary data for this work. Due to the gaining importance of SHIB crypto due to its multiplication of value from its start, we have taken SHIB trend analysis as a case study with the use of a data set provided by Yahoo Finance API. The novelty of the proposed system lies largely on the combination of wavelet and deep learning system to improve the performance measures. The proposed systems evaluates the wavelet transform coefficients of the historic data available from the Yahoo Finance API and feed the wavelet transform values to the input of Bi-LSTM network to predict the wavelet transform coefficients for the next 16 days. Finally an inverse wavelet transform is used to get the price of the crypto currency for next 16 days. The results obtained shows that the model was able to estimate the trend in crypto price. The trend analysis shows whether there is an increase or decrease of prices in upcoming days and this could be useful to invest or to withdraw the funds to improve the returns. The proposed scheme resulted in lower Mean Square Error (MSE), Mean Average Error (MAE) and Root Mean Square Error (RMSE) compared with the currently available methods. The method could be useful for trend analysis of other crypto currencies as well. Though it is practically impossible to get the trend analysis from the price and volume history alone, the proposed scheme gives a reasonable estimate of the trend of the crypto price. The work may be extended in future to applications like prediction of agricultural product price, variation in climatic conditions, and availability of certain agricultural products like Onion in the market.

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1. Introduction

Cryptocurrency has gained popularity in the recent past largely due to the profit given to early investors of bitcoin; based on the success of bitcoin, many alt-coins have been introduced into the market, and many of them have multiplied their values several folds in a short span of time. At the same time, cryptocurrency prices have been volatile and depend on many factors like comments by celebrities, policies of various countries, tweets, fraudulent activities and hype over social media. Profit and loss in such a volatile market are largely decided by the time of investment and withdrawals. Unfortunately, there is no proven mechanism that delivers 100% accuracy to predict a pump and dump in the market. Several methods have been used for the prediction of crypto prices.

Recently several researchers reported the use of artificial intelligence for the prediction of time series in many applications like agriculture, stock markets, health care, machine fault prediction and crypto prices. The ability of AI-based schemes to process large amounts of data and intelligent feature selection capability [13] have proved their strength in time series prediction applications, including cryptocurrency price prediction or trend analysis. Shiba inu token has gained considerable attention as an alt-coin. In this paper, we use this token for crypto price prediction. The algorithm can also be applied to any other coin by feeding the input data of the coin of interest.

Wavelet transforms are known for their ability to analyze non-stationary signals. Wavelet transforms split signals to high and low-frequency bands in an orthogonal manner. This helps to reduce the correlation between individual bands. This property of wavelets can be effectively used to extract features from time series data, and these uncorrelated features could be effectively used to attain better performance in AI systems. This paper explores a combination of AI and wavelet transforms to predict the trend of cryptocurrency price by using features extracted by wavelet transforms fed as input to the AI system. The model presented could be used for other applications demanding the prediction of non-stationary time series.

This paper is organized into nine sections; section 1 provides a brief introduction presenting the need of the system, novelty and an outline of the system. Section 2,3, and 4 presents a brief introduction to the technology used - Wavelets, Bi directional LSTM and various performance measure used to quantify the effectiveness of the proposed system, respectively. Section 5 presents an overall summary of previous works while section 6 provide a comprehensive description of the system used. Section 7 presents the results and comparison of the system with existing methods. Section 8 presents a conclusion followed by a reference section.

2. Wavelets an overview

Wavelet transforms are widely used for applications requiring the analysis of non-stationary signals. Wavelet transforms are defined by a set of filter banks with a scaling and wavelet filter, which divides the signal into dyadic bands [1],[2]. In this work, an un-decimated wavelet transform is used to decompose the signal into a high-frequency and low-frequency band in one level of transform. The low-frequency band is again passed through the wavelet transform step to get the second-level approximation (low frequency) and detail (high frequency) signals. The approximation signal obtained from the second level is again passed through the filter banks to get third-level coefficients. In Fig. 1, a schematic diagram of a three-level un-decimated wavelet transform is shown. The coefficients $h(n)$ and $g(n)$ represents the low pass and high pass filters for a given mother wavelet. CA1, CA2 and CA3 stand for approximation (low frequency) coefficients for levels 1, 2 and 3, respectively, while CD1, CD2 and CD3 stand for detail (high frequency) coefficients for levels 1, 2 and 3, respectively. In this work, Daubechies wavelet transform is used to obtain the coefficients. The wavelet transforms are computed by eq. 1. The function ψ in eq. 2 represents the mother wavelet function. A detailed study about wavelets can be found in [10].

$$T(m, n) = \int_{-\infty}^{\infty} x(t) \psi_{m,n}(t) dt \quad (1)$$

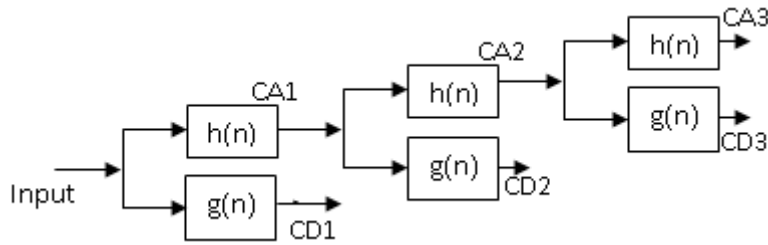


Fig. 1. Three-level wavelet transform structure (un-decimated scheme).

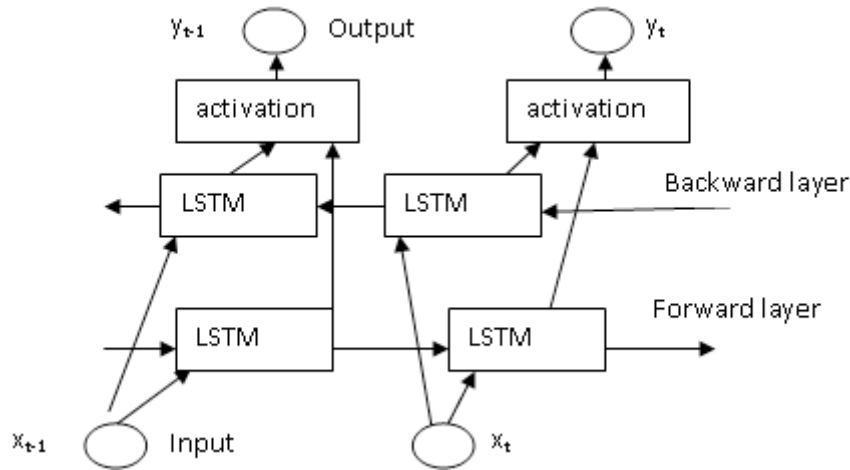


Fig. 2. Structure of Bi-LSTM network.

$$\text{where } \psi_{m,n}(t) = \frac{1}{\sqrt{2^j}} \psi\left(\frac{t - k2^j}{2^j}\right) \quad (2)$$

3. Bi directional LSTM an overview

Bidirectional Long Short Term Memory (Bi-LSTM) is a special neural network structure where information can be used in both directions – backward or future to past and forward past to future [12].

In Fig. 2, the inputs are fed to two LSTM layers (one for the forward and the other for the backward). The LSTM layers feed the activation layers to deliver the outputs. The network structure is scalable for any number of input and output variables [12].

4. Performance Measures

4.1. Mean Absolute Error (MAE)

MAE is a common metric used in machine learning and regression analysis to evaluate the performance of a predictive model. MAE measures the average absolute difference between the actual values (or ground truth) and the predicted values produced by the model. It provides a straightforward way to quantify the model's accuracy. MAE is calculated according to eq. 3

$$MAE = \frac{1}{n} \sum_{i=1}^n |actual_i - predicted_i| \quad (3)$$

where:

n is the total number of data points.

$actual_i$ is the actual value for the i -th data point.

$predicted_i$ is the predicted value for the i -th data point.

4.2. Mean Squared Error (MSE)

MSE calculates the average squared difference between the predicted values and the actual values. It penalizes larger errors more severely than MAE and is useful for understanding the variance of errors. MSE is calculated according to eq. 4

$$MSE = \frac{1}{n} \sum_{i=1}^n |actual_i - predicted_i|^2 \quad (4)$$

where:

n is the total number of data points.

$actual_i$ is the actual value for the i -th data point.

$predicted_i$ is the predicted value for the i -th data point.

4.3. Root Mean Squared Error (RMSE)

RMSE is the square root of the MSE given in eq.4. It provides an interpretable measure of the average error in the same units as the target variable.

Other performance measures like R-squared (R^2) Score are not suited for the problem since the prediction can not be done by a linear regression model due to the complexity of the crypto price dependency on historic data.

5. Literature Survey

Many research works have been reported for predicting future values of time series data. Some examples of time series data include the prediction of weather to decide various processes in agriculture prediction of natural calamities like cyclones, earthquakes etc. Prediction of financial data such as cryptocurrency price is a highly challenging task; the dependency of cryptocurrency on many factors from time to time makes a perfect prediction system an ideal system. The artificial intelligence-based scheme is gaining a lot of attention in the prediction of cryptocurrency prediction due to its ability to model complex problems to produce tangible results. This section provides a summary of work reported in the field of crypto price prediction.

In [5], Laxmi et al. proposed an implementation using recurrent neural network (RNN) and a Long Short Term Memory (LSTM) to predict bitcoin price from about 37 characteristics of it from "bitcoin.com". This data has been collected since July 2010. In this work, a regression line is obtained to get the best-fit curve for the bitcoin price with the help of the LSTM network designed using the python library 'Keras'[11]. A three-layer bi-directional LSTM architecture with dropout layers in between was used for system implementation in the work reported in [5]. However this model used a basic approach and the model was prone to false prediction. A method based on LSTM, Gated Recurrent Network (GRU) and Convolutional Neural Network (CNN) and an API is reported in [6] to predict bitcoin price for the next day. Data from [4] was used in [6], and the Keras library was used to develop the models with three hidden layers and a dense layer. Fifty epochs with a batch size of 8 are used for training the network, and it is reported that the GRU model predicted the values with an RMSE of 0.014 and with an accuracy of -0.0028. The work suggests use of better deep learning scheme to improve the performance further.

In [7], an Auto-Regressive Integrated Moving Average (ARIMA) model was proposed and found to provide a lower Mean Absolute Error (MAE) compared with LSTM and GRU for data obtained from CryptoCompare (<https://min-api.cryptocompare.com>). ARIMA model with an 80:20 train split model was reported with an absolute error close

Table 1. Summary of loss for various coefficients during the training phase

Parameter	Coefficients (Values after a minimum of 5 epochs considered)					
	cA3	cD3	cA2	cD2	cA1	cD1
Maximum training loss	0.0159	0.0046	0.0099	0.003	0.0024	0.0033
Minimum training loss	9.93E-04	5.89E-04	1.30E-03	5.25E-04	9.00E-04	3.10E-03
Training loss converges approximately to	9.58E-04	0.001	2.50E-03	5.30E-04	9.20E-04	3.10E-03
Maximum validation loss	0.1185	0.0704	0.0592	0.0517	0.065	0.038
Minimum validation loss	0.087	0.0215	0.00461	0.0169	0.021	0.0101
Validation loss converges approximately to	0.1	0.035	0.046	0.023	0.06	0.0101

to 0.02294 for the bitcoin price, which is higher compared to the bitcoin price. This work suggests incorporation of more features or complex network structure for further improvement. This work rely primarily on the assumption that the crypto price is a stationary time series. However in practise they belong to a non-stationary time series.

A method based on RNN, LSTM, GRU and news on social media platforms like Reddit and Twitter was proposed in [3] and achieved a minimum MAE of 1016 and MSE of 42147659 with LSTM. in this work the authors use scaling and reshaping as pre-processing steps for the historic data of bitcoin obtained from Coin Base, Coin Market Cap and Coin Desk along with the sentiment analysis from Twitter (currently known as X) and Reddit. Though the model was able to predict the price to some extent, the MSE value obtained was higher. Another scheme of comparison was done in [9] among RNN, LSTM, GRU and Bidirectional GRU(Bi-GRU) Bi-GRU gave better performance among these, Bi-LSTM achieved RMSE, MAE and R2 of 981.18,540.81 and 0.9926 respectively.

$$y = \frac{e^x - e^x}{e^x + e^x} \quad (5)$$

A seven-day prediction scheme was proposed in [8] for the cryptocurrency Lite-coin and Monero with Cauchy Exploration Strategy Beetle Antenna Search- Adaptive Neuro-Fuzzy Inference System (CESBAS-ANFIS) scheme. This approach uses hybrid machine learning and particle swarm intelligence to predict crypto prices. Training and testing with data from 24-8-2016 to 23-02-2020 for Litecoin and data from 30-01-2015 to 23-02-2020 for Monero were used in [8] and results obtained were compared with LSTM and GRU, and a marginal improve in MSE, RMSE, MAE and MAPE was reported.

Wavelets can be used to separate out uncorrelated segments in a signal. In [1] various applications of wavelets are mentioned, and the use of wavelet transforms to extract features to feed AI systems was mentioned to provide better results from AI when the signals having non-stationary behaviour need to be predicted or classified. Cryptocurrency prices exhibit non-stationary characteristics, and hence wavelet-based AI schemes may provide tangible solutions to cryptocurrency price production.

Based on the literature review conducted, there is a scope for system improvement if a preprocessing of data to help feature extraction is done, and the preprocessed data could be used as input to the AI system; this research work is derived from this hypothesis. Choice of wavelet is further justified by the non stationary behaviour of the input data such as volume and price. Wavelets are a proven method to combat effect of non stationarity in data processing [1].

6. System description

In the proposed work, we used data for the ‘Shiba Inu’ cryptocurrency (SHIB) from Yahoo Finance through the Yahoo Finance API with Python code. 760 days of data were read from the API to build the model and to predict the price trend (closing price) for the next 16 days.

6.1. Data pre-processing

The data provided daily of the crypto price Open, High, Low, Close, Adj Close and Volume for a cryptocurrency. To generate a good number of time series, the data samples are made into slices of length 128. The number of samples is fixed as a power of 2 for consistency in wavelet transform calculation. For predicting 16 future values for the last 128

Table 2. Performance comparison with CA3 only model with that consider all level coefficients

Date	Prediction by considering all levels of coefficients	Prediction by considering only CA3	Date	Prediction by considering all levels of coefficients	Prediction by considering only CA3
07-10-2022	0.00001078	0.00001090	21-09-2022	0.00001180	0.00001168
08-10-2022	0.00001075	0.00001088	22-09-2022	0.00001231	0.00001167
09-10-2022	0.00001073	0.00001087	23-09-2022	0.00001190	0.00001169
10-10-2022	0.00001079	0.00001087	24-09-2022	0.00001179	0.00001172
11-10-2022	0.00001076	0.00001086	25-09-2022	0.00001190	0.00001176
12-10-2022	0.00001088	0.00001086	26-09-2022	0.00001207	0.00001180
13-10-2022	0.00001084	0.00001087	27-09-2022	0.00001179	0.00001184
14-10-2022	0.00001102	0.00001089	28-09-2022	0.00001169	0.00001187
15-10-2022	0.00001090	0.00001091	29-09-2022	0.00001157	0.00001189
16-10-2022	0.00001114	0.00001095	30-09-2022	0.00001202	0.00001191
17-10-2022	0.00001112	0.00001100	01-10-2022	0.00001207	0.00001191
18-10-2022	0.00001110	0.00001105	02-10-2022	0.00001189	0.00001190
19-10-2022	0.00001132	0.00001110	03-10-2022	0.00001195	0.00001188
20-10-2022	0.00001119	0.00001115	04-10-2022	0.00001184	0.00001187
21-10-2022	0.00001098	0.00001119	05-10-2022	0.00001164	0.00001186
22-10-2022	0.00001101	0.00001123	06-10-2022	0.00001159	0.00001187

days has to be given as input to the system; 512 days were used for training data, thereby $512-128=384$ training sample data. The remaining $(760-512=242)$ days data was used for validation, providing $242-128=114$ validation sample data. The data slicing operations to generate the time series samples for training and validation and the remaining parts of the system are implemented using Python code and executed in 'Google Colab'.

Fig. 6 represents the block schematic of the proposed system. A detailed description of the system is provided in this section.

Three levels of un-decimated stationary wavelet transform with Daubechies 3 wavelet are used to obtain the wavelet transform coefficients. Un-decimated transform provides consistency in several samples across the network. The wavelet transform coefficients are calculated for all variables (open, high, low, close, adjacent close and volume). As the transform values at different levels show wide variations in their magnitude, scaling the transform coefficient is done before passing it as the input to the AI network. The Min-max scaling is applied to all coefficients to improve the model accuracy and to avoid the effect of wide variation in approximation and detail coefficients across the levels.

Table 3. A comparison between the actual and predicted price of SHIB on 20-9-2022

Date	Prediction by considering all levels of coefficients	Prediction by considering only CA3	Date	Prediction by considering all levels of coefficients	Prediction by considering only CA3
07-10-2022	0.00001078	0.00001090	21-09-2022	0.00001180	0.00001168
08-10-2022	0.00001075	0.00001088	22-09-2022	0.00001231	0.00001167
09-10-2022	0.00001073	0.00001087	23-09-2022	0.00001190	0.00001169
10-10-2022	0.00001079	0.00001087	24-09-2022	0.00001179	0.00001172
11-10-2022	0.00001076	0.00001086	25-09-2022	0.00001190	0.00001176
12-10-2022	0.00001088	0.00001086	26-09-2022	0.00001207	0.00001180
13-10-2022	0.00001084	0.00001087	27-09-2022	0.00001179	0.00001184
14-10-2022	0.00001102	0.00001089	28-09-2022	0.00001169	0.00001187
15-10-2022	0.00001090	0.00001091	29-09-2022	0.00001157	0.00001189
16-10-2022	0.00001114	0.00001095	30-09-2022	0.00001202	0.00001191
17-10-2022	0.00001112	0.00001100	01-10-2022	0.00001207	0.00001191
18-10-2022	0.00001110	0.00001105	02-10-2022	0.00001189	0.00001190
19-10-2022	0.00001132	0.00001110	03-10-2022	0.00001195	0.00001188
20-10-2022	0.00001119	0.00001115	04-10-2022	0.00001184	0.00001187
21-10-2022	0.00001098	0.00001119	05-10-2022	0.00001164	0.00001186
22-10-2022	0.00001101	0.00001123	06-10-2022	0.00001159	0.00001187

6.2. Model architecture

As we used an un-decimated wavelet coefficient to ensure consistency of the number of features across levels, the network size used was also proportionally high. A Bi-directional LSTM model with 700 layers processes the input wavelet transform coefficients. The LSTM layer is followed by a dense layer of size 90 with the tanh activation function. The dense layer feeds the next layer, a Bi-directional LSTM model with 250 layers. This layer is followed by three dense layers of size 70, 65 and 60 with tanh activation. A dropout layer with a dropout of 0.4 is used before feeding to the final dense layer with a size compatible with the output dimension required.

The tanh activation function is defined by the rule given in eq. 5.

The model has compiled with ‘adam’ optimizer with parameters loss as ‘mean squared error’ with ‘accuracy’ as a metric. The model is built with a batch size of 256 and a buffer size 128 and fitted with training data with 150 epochs; each has 100 steps and 50 validation steps. The model is trained with an early stopping with patience 50 and

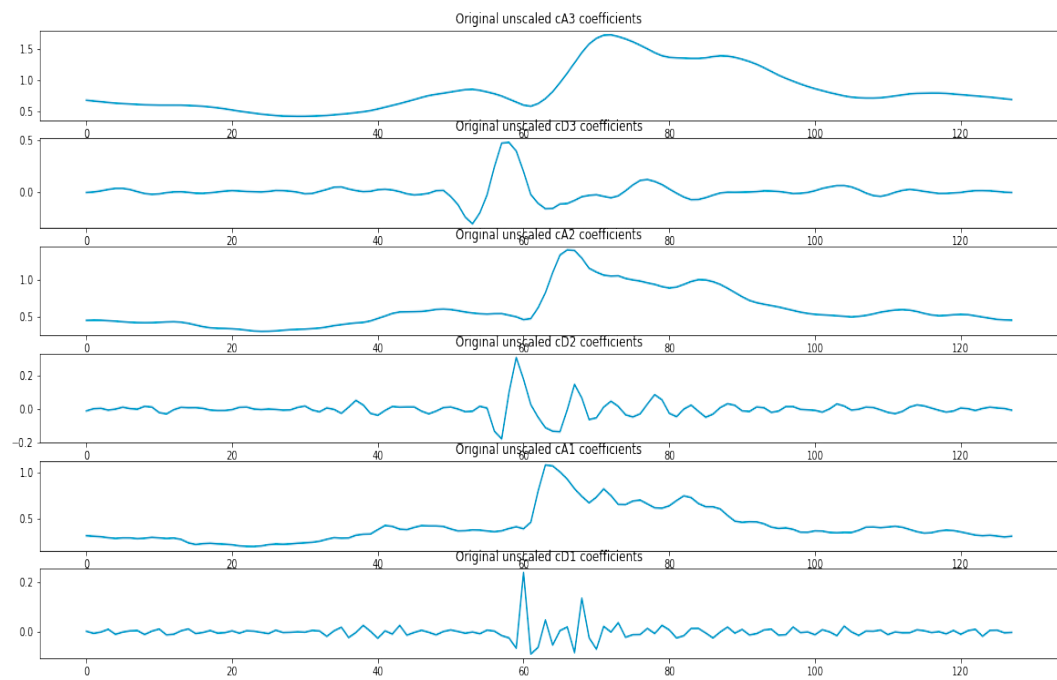


Fig. 3. Three level wavelet coefficients - a sample.

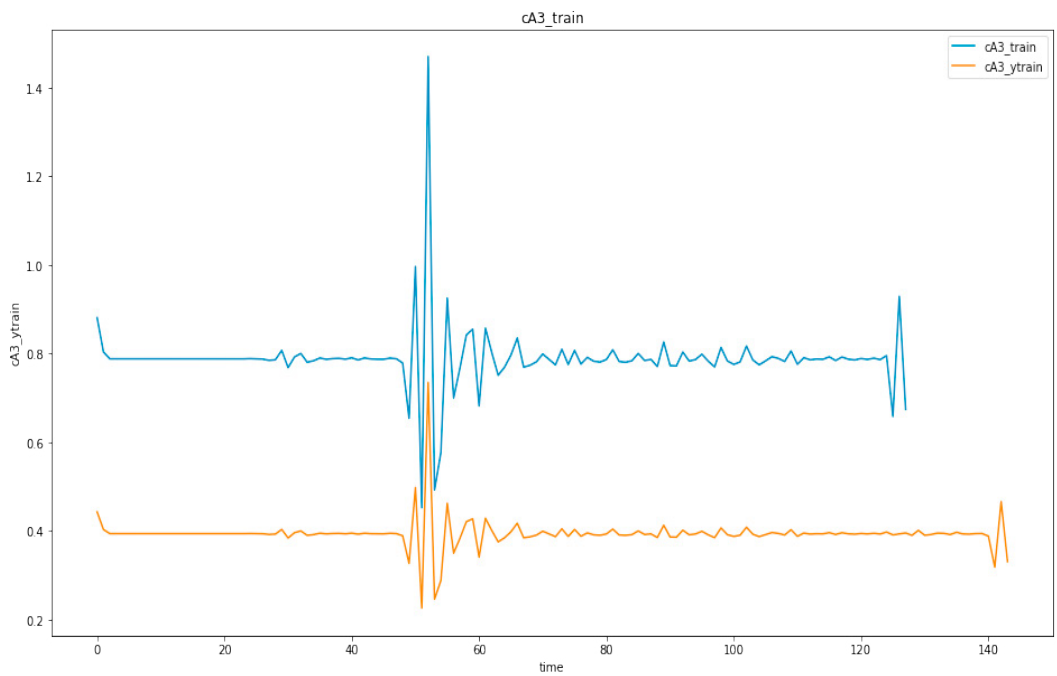


Fig. 4. Third level approximation coefficients - a sample.

monitoring validation loss and minimum delta=0 and the best only model is selected and saved for predicting future values.

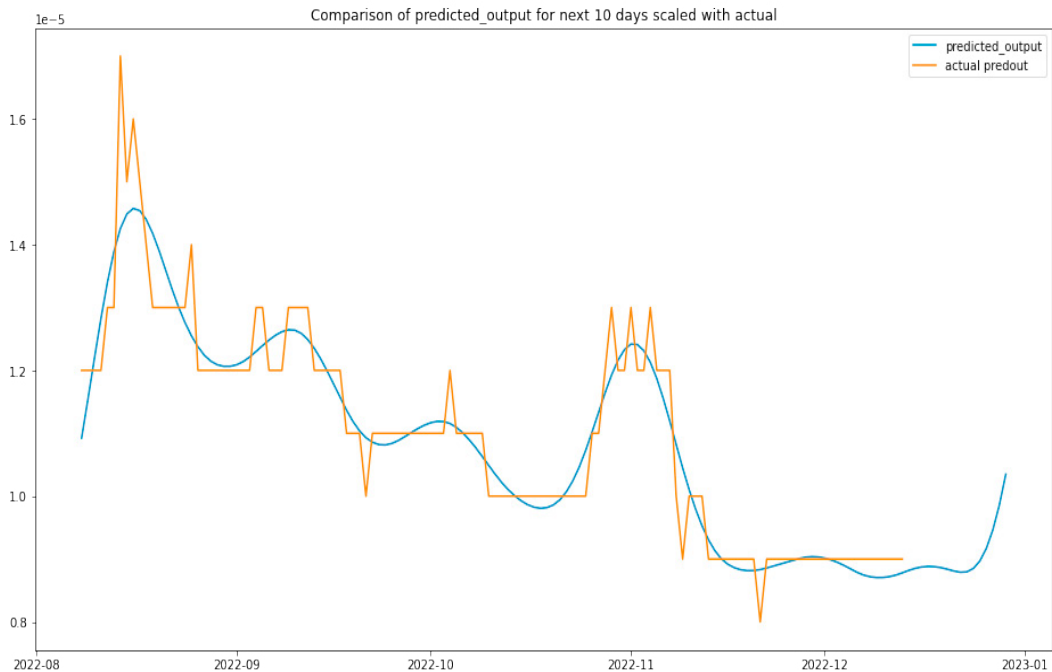


Fig. 5. Trend prediction of SHIB, graph predicted on 13 December 2022.

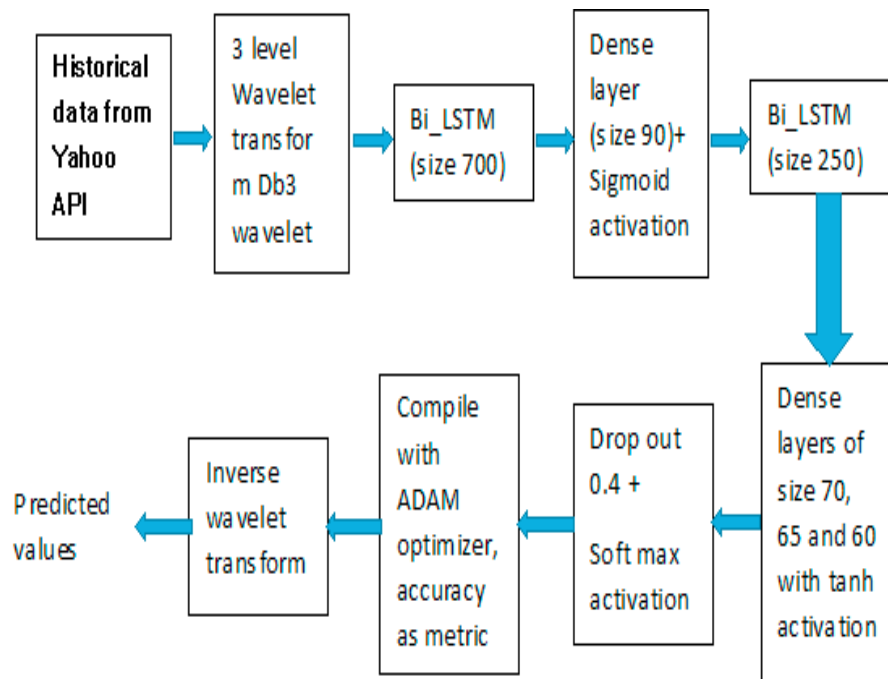


Fig. 6. System block diagram.

Table 4. Performance measures and comparison with previous results

Performance measure	Proposed method	Author1 [7]	Author2 [6]
MAE	0.00000094	0.02294	–
MSE	0.000000000001	–	0.000196
RMSE	0.00000097	–	0.014

7. Results and discussions

Fig. 3 shows the wavelet transform coefficients of the time series data. The notation cA3, cA2 and cA1 shows the Approximation coefficient of level 3, 2 and 1 of wavelet transform, respectively; while cD3, cD2 and cD1 is detail coefficients of the same.

Fig. 4 shows the values of the cA3 train are scaled up by two only to show the similarity. However, the data samples of training input are a time-delayed version of output samples. To differentiate between the plots, CA3 train values are scaled up by a factor of two only to plot the graph. The cA3train and cA3_ytrain show similar characteristics, except for the number of samples in each.

The network has been trained, and results are tested by considering all levels of wavelet coefficients. The process is repeated with only first-level approximation coefficients as well. Table 1 shows the changes are marginal between the two models. Compared with the computational complexity, and our major aim was to analyze the market trend rather than predicting exact values, the model with approximation coefficients only is considered for further analysis. Table 2 shows the outcome of predictive models considering all levels coefficients and that with only one level coefficient (CA3) only. The prediction values differ marginally only, and the aim of the prediction is the trend analysis due to the reduction in computational time for training, validation and prediction system with only the CA3 coefficient used for trend analysis further in this work.

Fig. 5 and Table3 show the trend analysis results for the coin in different time frames. The coin price follows the trend reasonably, showing the effectiveness of the predictor system.

Table 3 shows that the predictor algorithm predicted the trend – there will not be much variation in price in the coming days on 20 Sept 2022 to 6 October 2022. Except for 4 October 2022, the trend predicted was correct.

Table 4 presents the performance measure of the proposed system with that quoted in literature. It is evident that the ability of wavelet transform to handle the non stationary behavior of historic data plays an important role to reduce the errors and improving the effectiveness of the predictor compared with the deep learning systems quoted in the literature.

8. Conclusion

Wavelet-based feature extraction separates the crypto price data into uncorrelated values. This uncorrelated data improves the training and prediction capability of AI schemes for better time series prediction. The performance of predictive system shows an improvement over existing system. MSE shows a significant reduction (from 0.02294 to 0.00000094) while MSE also has been reduced significantly from 0.000196 to 0.000000000001. The analysis can be used as one indicator to plan the investment in crypto currency to decide on buying and selling. However, still, there is a scope for improvement of the scheme. Perhaps the inclusion of more features related to news related to crypto, days of the week etc may be considered in future work. It's important to note that crypto price is highly volatile, and predictions are often speculative. Many experts caution against relying solely on predictions and advise diversifying investments and conducting thorough research before making any decisions.

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